EXPERIMENT 06 **241401038**

Automating Business Processes Using Deluge Scripting and Workflows, with Application Testing and Debugging

Patient Appointment Booking Form — Workflow Automation Use Case

1. **Aim**

   To automate a key business process of the Patient Appointment & Scheduling Management application — sending an appointment confirmation email the moment a patient submits the Patient Profile form — using a Zoho Creator workflow and a Deluge script, and to test and debug the automation end-to-end with real form submissions. This removes the need for front-desk staff to manually notify every patient, ensuring each submitted appointment request is acknowledged immediately and consistently.

2. **Procedure**

   Step 1: Start a new workflow — From the application's Workflow section, a new workflow was created on the Patient Profile form.

   Step 2: Define the trigger — The Record Event was set to Created or Edited and the Form Event (When to trigger workflow) was set to Successful form submission, so the automation fires only after a valid, saved submission.

   Step 3: Name the workflow — The workflow was named Send Appointment Confirmation Email and saved using Create Workflow.

   Step 4: Add a new action — In the Add New Action dialog, Run this action was left on Always, and Deluge script was chosen as the Action type so custom logic could compose and send the email.

   Step 5: Write the Deluge script — A sendmail script was written in the Deluge editor: it sets from to zoho.loginuserid, to to input.email_address, builds a Subject of “Appointment Booking Confirmation – Patient Appointment & Scheduling Management”, and concatenates a message with the patient's name, age, gender, department, preferred date and time, existing conditions, and symptoms pulled directly from the form's input fields.

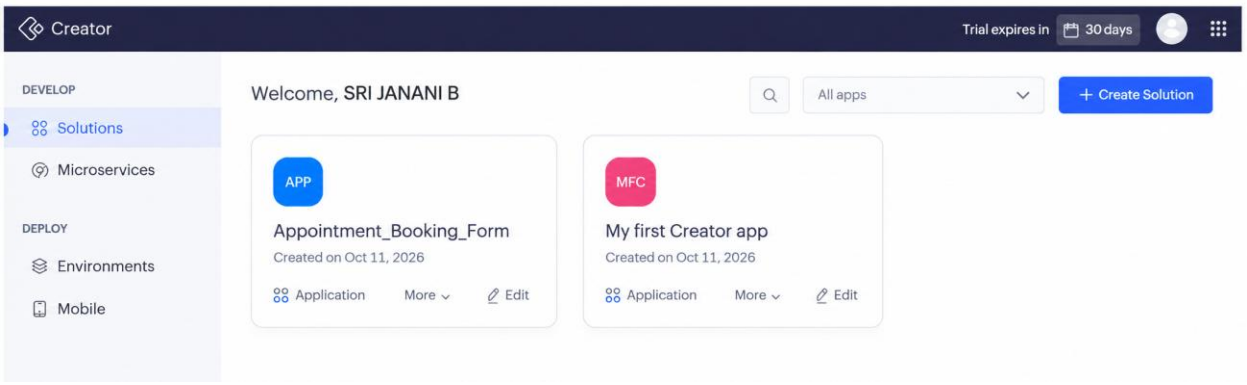
   Step 6: Save and review the action — The script was saved with Update, and the workflow summary confirmed the single Deluge script action was attached under Patient Profile › Created › Successful form submission › Send Appointment Confirmation Email, set to Execute without condition.

   Step 7: Test with a live submission — A new Patient Profile record was filled in through the live form — Department: Dermatology, Existing Conditions: Allergy to Penicillin, Diabetes, Heart Disease, Asthma, Preferred Appointment Date: 04-Oct-2026, Consent For Treatment checked — and submitted to trigger the workflow.

   Step 8: Debug and verify the output — The patient's inbox was checked immediately after submission to confirm the automation ran correctly: the confirmation email arrived with the correct dynamic values (name, age, gender, department, date, time, and conditions), confirming the Deluge script executed without errors.

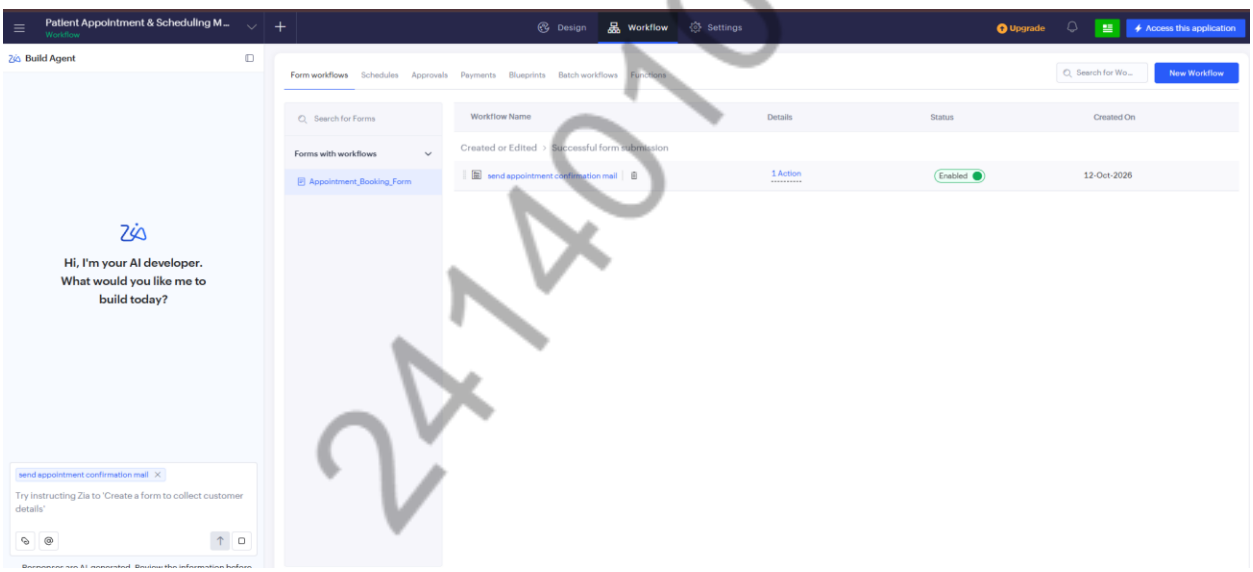
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3.1 Application in Zoho Creator



The Appointment_Booking_Form application listed on the Zoho Creator home page, opened in Edit mode to build the workflows.

3.2 First workflow created — trigger configuration



The Form workflows tab shows Send appointment confirmation mail attached to Appointment_Booking_Form under Created or Edited › Successful form submission, with 1 action and status Enabled.

3.3 Deluge script — confirmation email

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The Deluge script action (Execute without condition). The script converts the date, time and age to text, builds the conditions list and sends the confirmation with sendmail.

Full script used:

```
patientName = input.Patient_namePatient_name.first_name + " " +
input.Patient_namePatient_name.last_name;

apptDate = input.Preferred_Appointment_Date.toString("dd-MMM-yyyy");

apptTime = "Not specified";
if(input.Preferred_Appointment_Time != null)
{
    apptTime = input.Preferred_Appointment_Time.toString();
}

ageText = input.Age.toString();

conditions = "None";
if(input.Existing_Conditions != null && input.Existing_Conditions.size() > 0)
{
    conditions = input.Existing_Conditions.toString(", ");
}

symptoms = ifnull(input.Symptoms_Description, "-");

info "Workflow started for: " + patientName + " | Email: " + input.Email_Address1;

if(input.Email_Address1 != null && input.Email_Address1 != "")
{
    msg = "<escape></escape>

Dear " + patientName + ",

Your appointment request has been received.

";

    msg = msg + "
- Age: " + ageText + "
```

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```
- Gender: " + input.Gender + "
- Department: " + input.Department + "
";

    msg = msg + "
- Preferred Date: " + apptDate + "
- Preferred Time: " + apptTime + "
";

    msg = msg + "
- Existing Conditions: " + conditions + "
- Symptoms: " + symptoms + "

";

    msg = msg + "

Thank you,
Hospital Front Desk

";

    sendmail
    [
        from : zoho.loginuserid
        to : input.Email_Address1
        subject : "Appointment Booking Confirmation - " + input.Department
        message : msg
    ];

    info "Confirmation email sent to " + input.Email_Address1;
}
else
{
    info "No email address given, so no email was sent";
}
```

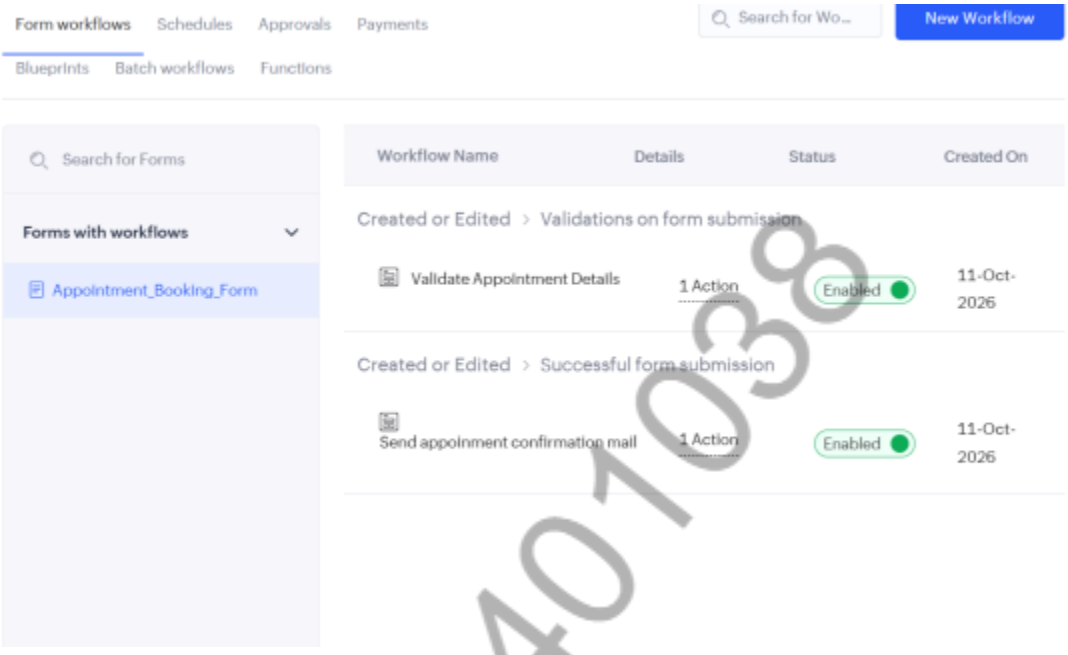
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3.4 Deluge script — validation on form submission



The breadcrumb Appointment_Booking_Form › Created or Edited › Validations on form submission › Validate Appointment Details. The script rejects an invalid age or a past appointment date with an alert and cancel submit , and logs a message with info when validation passes.

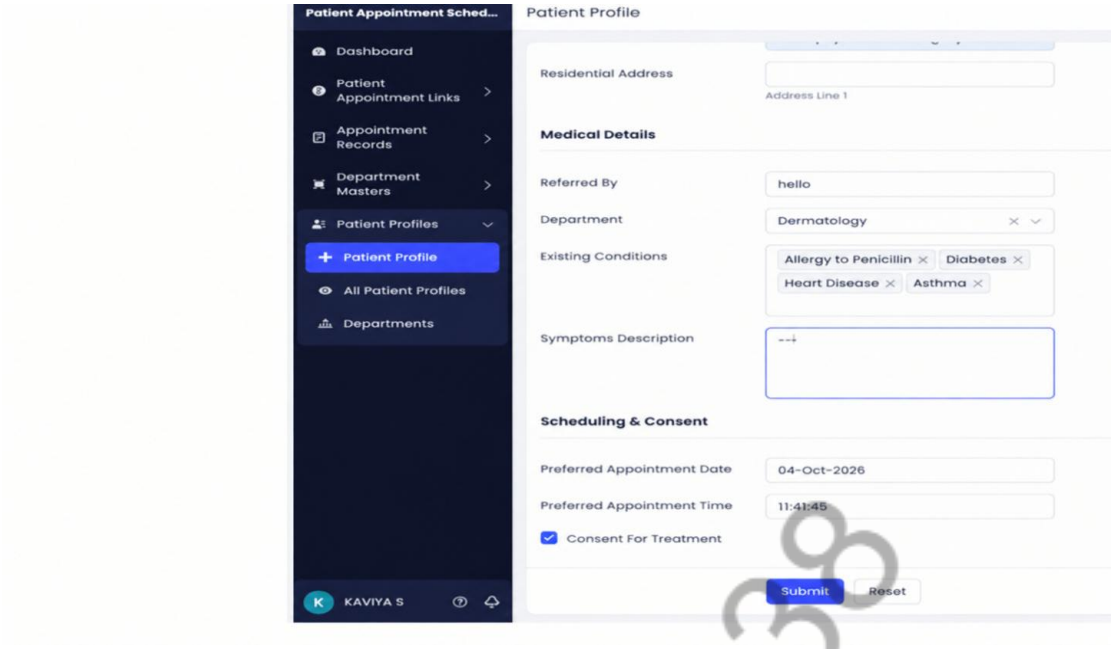
3.5 Saved workflows — summary view



Both workflows on Appointment_Booking_Form are Enabled: Validate Appointment Details (Validations on form submission) runs first and can stop the record, and Send appointment confirmation mail (Successful form submission) runs only after the record is saved.

3.6 Testing — submitting a live record

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A real Patient Profile record was filled in and submitted through the live form (Department: Dermatology, four Existing Conditions selected, Preferred Appointment Date 04-Oct-2026, Consent checked) specifically to trigger the Send Appointment Confirmation Email workflow and observe its behavior.

3.7 Debugging output — confirmation email received



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The received email confirms the automation ran successfully: the subject and message match the Deluge script exactly, and every dynamic value (Patient Name: HEMAPRIYA, Age: 19, Gender: Female, Department: Dermatology, date, time, and existing conditions) was pulled correctly from the submitted record, verifying the workflow end-to-end.

4. Final Result

A complete workflow was built on the Patient Profile form that fires automatically on successful submission, using a Deluge sendmail script to generate a personalised confirmation email with no manual intervention from the front desk.

The automation was tested with a live form submission and debugged by inspecting the resulting email: all dynamic fields rendered correctly, confirming the script's field references and string concatenation were error-free.

The exercise met its objective: a real business process — notifying patients that their appointment request was received — is now fully automated through low-code workflow and Deluge scripting, and the testing step confirmed the automation behaves reliably end-to-end

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